

Experiment-2

- Objective-1:Write a program function in C++ to swap two numbers using call by value, call by reference and call by address.

❖ Call by Value Code:

```
#include <bits/stdc++.h>
using namespace std;

class SwapDemo {
public:

    void swapByValue(int a, int b) {
        int t = a;
        a = b;
        b = t;
        cout << "Inside Call by Value: a = " << a << ", b = " << b << endl;
    }

};

int main() {
    SwapDemo x;
    int a, b;

    cout << "Enter two numbers: ";
    cin >> a >> b;

    x.swapByValue(a, b);
    cout << "After Call by Value: a = " << a << ", b = " << b << endl;

    return 0;
}
```

Output:

```
PS C:\Dishank\C++ codes> cd "c:\Dishank\CO  
OPS Lab\Experiments\Exp2\" ; if ($?) { g+  
?) { .\swapbyvalue }  
• Enter two numbers: 5 6  
  Inside Call by Value: a = 6, b = 5  
  After Call by Value: a = 5, b = 6  
• PS C:\Dishank\C++ codes\All the college u
```



Call by reference Code:

```
#include <bits/stdc++.h>
using namespace std;

class SwapDemo {
public:
    void swapByReference(int &a, int &b) {
        int t = a;
        a = b;
        b = t;
    }
};

int main() {
    SwapDemo x;
    int a, b;

    cout << "Enter two numbers: ";
    cin >> a >> b;
    x.swapByReference(a, b);
    cout << "After Call by Reference: a = " << a << ", b = " << b << endl;

    return 0;
}
```

Output:

```
PS C:\Dishank\C++ codes> cd "c:\Dishank\C++ codes"
g++ swapbyreference.cpp -o swapbyreference } ; if ($?) { .\swapbyreference.exe
Enter two numbers: 4 5
After Call by Reference: a = 5, b = 4
PS C:\Dishank\C++ codes\All the college work>
```



Call by address Code:

```
#include <bits/stdc++.h>
using namespace std;

class SwapDemo {
public:
    void swapByAddress(int *a, int *b) {
        int t = *a;
        *a = *b;
        *b = t;
    }
};

int main() {
    SwapDemo x;
    int a, b;

    cout << "Enter two numbers: ";
    cin >> a >> b;
    x.swapByAddress(&a, &b);
    cout << "After Call by Address: a = " << a << ", b = " << b << endl;

    return 0;
}
```

Output:

```
PS C:\Dishank\C++ codes> cd "c:\Dishank\
p -o swapbyaddress } ; if ($?) { .\swapt
Enter two numbers: 4 5
After Call by Address: a = 5, b = 4
PS C:\Dishank\C++ codes\All the college
```

- Objective-2:Write a program function in C++ to increment a number by 1 using call by value, call by address and call by reference.

❖ Call by value Code:

```
#include <bits/stdc++.h>
using namespace std;

class incrementdemo{
public:
    void incrementbyvalue(int a){
        int t=a;
        a++;
        cout<<"Before increment: "<<t<<endl;
        cout<<"After increment: "<<a<<endl;
    }
};

int main(){
    incrementdemo x;
    int a;
    cin>>a;
    x.incrementbyvalue(a);
}
```

Output:

```
PS C:\Dishank\C++ codes> g++ incrementbyvalue.cpp -o incrementbyvalue }
5
Before increment: 5
After increment: 6
PS C:\Dishank\C++ codes\All
```

❖ Call by reference Code:

```
#include <bits/stdc++.h>
using namespace std;

class incrementdemo{
public:
    void incrementbyreference(int &a){
        int t=a;
        a++;
        cout<<"Before increment: "<<t<<endl;
        cout<<"After increment: "<<a<<endl;
    }
};

int main(){
    incrementdemo x;
    int a;
    cin>>a;
    x.incrementbyreference(a);
}
```

Output:

```
● PS C:\Dishank\C++ codes> cd "c:\Dishank\C++ codes"
enece.cpp -o incrementbyreferenece
6
Before increment: 6
After increment: 7
○ PS C:\Dishank\C++ codes\All the col
```

❖ Call by address Code:

```
#include <bits/stdc++.h>
using namespace std;

class incrementdemo{
public:
    void incrementbyvalue(int *a){
        int t=*a;
        (*a)++;
        cout<<"Before increment: "<<t<<endl;
        cout<<"After increment: "<<*a<<endl;
    }
};

int main(){
    incrementdemo x;
    int a;
    cin>>a;
    x.incrementbyvalue(&a);
}
```

Output:

```
● PS C:\Dishank\C++ codes> cd "C:\Dishank\C++ codes" & gcc ss.cpp -o incrementbyaddress }
70
Before increment: 70
After increment: 71
○ PS C:\Dishank\C++ codes\All the
```

- Objective-3: To write an inline keyword function in C++.

❖ Code:

```
#include <iostream>
using namespace std;

inline int square(int x)
{
    return x * x;
}

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    cout << "Square of " << num << " is " << square(num) << endl;

    return 0;
}
```

Output:

```
● PS C:\Dishank\C++ codes> cd
pp -o inlinefunction } ; if
Enter a number: 5
Square of 5 is 25
○ PS C:\Dishank\C++ codes\All
```